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**Theme** — *Science & Technology for Sustainable Future*

**Sub-Theme** — *Emerging Technologies*

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**Class** — *XII - Science*

**Subject** — *Information Technology (IT)*

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**Session** — *2026–2027*



# EXTINGO

*Autonomous Fire Detection & Suppression Robot*

## Project Logbook

Duration: 13 Days | 12 August 2026 – 24 August 2026

### Purpose of this Logbook

This logbook records the day-wise design, build, and testing process followed while developing Extingo, from problem definition through to the final integrated fire-simulation demo. Each entry lists the day's objective, the work carried out, observations made, and any challenges encountered along with how they were resolved — giving a transparent, chronological account of how the robot's scanning, detection, alerting, and suppression subsystems were built up one at a time and finally merged into a single working system.

## Day 1 — 12 August 2026 | *Problem Definition & Planning*

|                             |                                                                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Define the problem statement and finalize project scope.                                                                                                               |
| <b>Work Carried Out</b>     | Researched existing fire-detection systems, listed required sensors and actuators, sketched an initial block diagram, and finalized the 13-item component list.        |
| <b>Observations</b>         | Confirmed that a priority-based sensor fusion (IR > Heat > Smoke) is the most practical basis for early, reliable detection.                                           |
| <b>Challenges &amp; Fix</b> | Undecided between a single combo GPS+GSM board or two modules; resolved to use a separate NEO-6M GPS and SIM800L GSM module, since no single board reliably does both. |

## Day 2 — 13 August 2026 | *Base Setup — Arduino Uno*

|                             |                                                                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Set up the development environment and verify base hardware.                                                                                                                             |
| <b>Work Carried Out</b>     | Installed Arduino IDE and required libraries (Servo, Wire, Adafruit_GFX/SSD1306, DHT, SoftwareSerial, TinyGPS++); ran a blink test; planned the regulated 5V power rail for all sensors. |
| <b>Observations</b>         | The Uno has only one hardware serial port, insufficient for GSM and GPS together — decided to use SoftwareSerial for one of them.                                                        |
| <b>Challenges &amp; Fix</b> | USB-serial driver not recognized on the board (CH340 clone); resolved by installing the correct driver.                                                                                  |

## Day 3 — 14 August 2026 | *I2C OLED Interfacing*

|                             |                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Interface the I2C OLED and display the boot message.                                                                                                                                  |
| <b>Work Carried Out</b>     | Wired SDA/SCL to A4/A5, ran an I2C scanner to confirm the display address, coded and tested the “Extingo – Ready” boot text, and built a reusable template for later sensor readouts. |
| <b>Observations</b>         | Display refresh was clean with negligible current draw, suitable for continuous use.                                                                                                  |
| <b>Challenges &amp; Fix</b> | Screen stayed blank initially due to a wrong assumed I2C address; fixed after running the scanner and updating the address in code.                                                   |

## Day 4 — 15 August 2026 | *Pan-Tilt Gimbal Assembly*

|                             |                                                                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Build the two-servo pan-tilt gimbal and verify sweep motion.                                                                                                         |
| <b>Work Carried Out</b>     | Mounted the pan servo as the base and the tilt servo perpendicular on top; tested 0–180° sweep on both axes; mounted the IR sensor bracket parallel to the tilt arm. |
| <b>Observations</b>         | Minor mechanical backlash at extreme angles; a 15 ms step delay gave smooth, jitter-free motion.                                                                     |
| <b>Challenges &amp; Fix</b> | Servos jittered under load from insufficient current on the Uno's 5V pin; fixed by powering servos from a separate 5V/2A supply with a common ground.                |

## Day 5 — 16 August 2026 | *IR Sensor & Scan Logic*

|                             |                                                                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Interface the IR flame sensor and implement the N/E/S/W scanning routine.                                                                                        |
| <b>Work Carried Out</b>     | Connected the IR digital output; wrote the scan loop mapping pan angles to North/East/South/West, pausing at each stop to read the sensor and log the direction. |
| <b>Observations</b>         | Reliable detection of a small flame at ~40 cm; occasional false triggers under direct sunlight.                                                                  |
| <b>Challenges &amp; Fix</b> | Sunlight interference caused false positives; added a 2-sample debounce/confirmation check before accepting a detection.                                         |

## Day 6 — 17 August 2026 | *DHT22 & Rate-of-Rise Logic*

|                             |                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Integrate the DHT22 heat sensor and begin the rate-of-rise algorithm.                                                                                             |
| <b>Work Carried Out</b>     | Wired the DHT22 with pull-up resistor; sampled temperature every 2 s; coded $\Delta T/\Delta t$ calculation; tested with a hot-air gun to simulate rapid heating. |
| <b>Observations</b>         | Algorithm correctly flagged a simulated rise of $\sim 2.3^\circ\text{C}/\text{sec}$ , matching the design threshold.                                              |
| <b>Challenges &amp; Fix</b> | Occasional failed reads due to DHT22's slow response; added a retry-and-error-check routine around each read.                                                     |

## Day 7 — 18 August 2026 | *MQ2 Sensor & Priority Fusion*

|                             |                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Integrate the MQ2 smoke sensor and finalize the 3-sensor priority logic.                                                                                                                        |
| <b>Work Carried Out</b>     | Connected MQ2 analog output; calibrated clean-air baseline and threshold; coded the $IR(1) > Heat(2) > Smoke(3)$ decision logic; validated with a candle test while logging all three readings. |
| <b>Observations</b>         | IR responded fastest (1–2 s), MQ2 lagged by several seconds — confirming the priority design was justified.                                                                                     |
| <b>Challenges &amp; Fix</b> | MQ2 needs a 24–48 hr burn-in for stable baseline readings; noted this as a pre-final-test requirement.                                                                                          |

## Day 8 — 19 August 2026 | *GSM Module — SMS & Call*

|                             |                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Interface the SIM800L GSM module for SMS and voice-call alerts.                                                                                                                       |
| <b>Work Carried Out</b>     | Wired SIM800L on SoftwareSerial with its own regulated supply; tested AT commands; sent a test SMS and placed a test call; drafted the SMS template with a placeholder for direction. |
| <b>Observations</b>         | SMS delivery took ~5–8 s; module reset unexpectedly during call attempts under load.                                                                                                  |
| <b>Challenges &amp; Fix</b> | Voltage sag during GSM transmit bursts caused resets; added a 1000 $\mu$ F capacitor across the module's supply rail, which stabilized it.                                            |

## Day 9 — 20 August 2026 | *GPS Module & Location Tagging*

|                             |                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Integrate the NEO-6M GPS module and merge coordinates into the SMS alert.                                                                                                  |
| <b>Work Carried Out</b>     | Wired GPS on a second SoftwareSerial port; parsed NMEA sentences with TinyGPS++; extracted latitude/longitude; combined them into the SMS message format; tested outdoors. |
| <b>Observations</b>         | Cold-start fix took ~45 s outdoors; no fix indoors, as expected for GPS.                                                                                                   |
| <b>Challenges &amp; Fix</b> | Running two SoftwareSerial ports (GSM + GPS) together caused data loss; reduced the GPS baud rate to 9600 and adjusted read timing to resolve it.                          |

## Day 10 — 21 August 2026 | L298N — Exhaust & Pump Driving

|                             |                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Set up the L298N motor driver for the exhaust fan and water pump.                                                                                                       |
| <b>Work Carried Out</b>     | Wired the pump to OUT1/OUT2 and the exhaust fan to OUT3/OUT4; tested digital ON/OFF and PWM speed control via ENA/ENB; verified pump flow at different PWM duty cycles. |
| <b>Observations</b>         | Clear difference in pump flow between 40% and 90% PWM duty; exhaust fan ran smoothly at a fixed speed.                                                                  |
| <b>Challenges &amp; Fix</b> | L298N ran hot under continuous pump load; fitted a small heatsink to keep it within safe operating temperature.                                                         |

## Day 11 — 22 August 2026 | *PIR Sensor & Pump Pressure Logic*

|                             |                                                                                                                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Integrate the PIR motion sensor and implement occupancy-based pump pressure control.                                                                                                                             |
| <b>Work Carried Out</b>     | Wired the PIR sensor; characterized its detection range and warm-up time; coded the logic — person present → low PWM, no person → high PWM; validated with a team member simulating presence during a mock test. |
| <b>Observations</b>         | PIR needed ~30 s warm-up after power-on for stable output; ~5 m range was adequate for room-scale testing.                                                                                                       |
| <b>Challenges &amp; Fix</b> | False triggers from a nearby moving curtain; corrected by adjusting the PIR's sensitivity potentiometer.                                                                                                         |

## Day 12 — 23 August 2026 | *ISD1820 Integration & Full Code Merge*

|                             |                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Integrate the ISD1820 sound module and merge all subsystems into a single sketch.                                                                                                                                                       |
| <b>Work Carried Out</b>     | Recorded the warning message on the ISD1820; wired it for loop-playback triggered from a digital pin; merged scanning, detection, GSM, GPS, PIR, pump, exhaust, siren and OLED code into one program, resolving pin/variable conflicts. |
| <b>Observations</b>         | Combined sketch used ~78% of the Uno's flash memory — within limits, but tight.                                                                                                                                                         |
| <b>Challenges &amp; Fix</b> | Random resets from SRAM overflow; fixed by moving string constants to flash with the F() macro and trimming global buffers.                                                                                                             |

## Day 13 — 24 August 2026 | *Final Integration Test & Demo*

|                             |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>            | Run full end-to-end testing, calibration, and a live fire-simulation demo.                                                                                                                                                                                                                                                                 |
| <b>Work Carried Out</b>     | Conducted a mock fire test: introduced a candle flame during scanning; verified gimbal hold, SMS + GPS + call sequence, PIR-based pump pressure switching, exhaust and siren activation, and 3-second OLED refresh; confirmed auto-reset after flame removal; tested the rate-of-rise SMS and “all clear” confirmation with a hot-air gun. |
| <b>Observations</b>         | End-to-end response time from flame detection to SMS delivery was ~9 s; all subsystems performed as designed, with the main detection loop and rate-of-rise watchdog running correctly in parallel.                                                                                                                                        |
| <b>Challenges &amp; Fix</b> | Slight timing overlap between the siren loop and the call command; added a simple priority queue for GSM commands to fix it. Project is functionally complete and ready for evaluation.                                                                                                                                                    |

## Summary

Across 13 days, Extingo progressed from a paper design to a fully integrated robot: Days 1–2 covered planning and base setup, Days 3–5 built the display and scanning mechanism, Days 6–7 completed the fire-sensing trio and its priority logic, Days 8–9 added communication and location, Days 10–11 brought suppression hardware online, and Days 12–13 merged every subsystem and validated it end-to-end.

thank you